

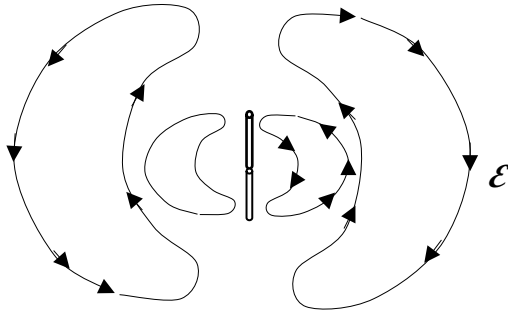
Notes #7 Read Chapter 7 in Balanis
Antenna duality

Short Dipole

$$\mathbf{E} = I l j \eta k \frac{e^{-jkr}}{4\pi r} \sin \theta \hat{\boldsymbol{\theta}}$$

$$\mathbf{H} = \underbrace{I l j k}_{\doteq P} \frac{e^{-jkr}}{4\pi r} \sin \theta \hat{\boldsymbol{\phi}}$$

$\doteq P$ (electric dipole moment)



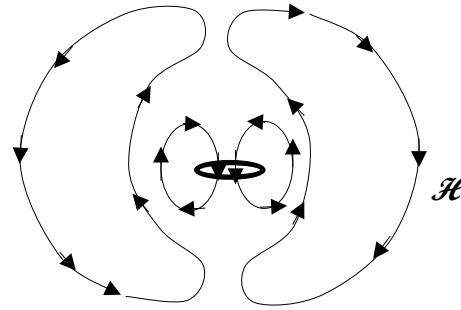
length of dipole = l

Small circular loop

$$\mathbf{E} = I(\pi a^2) \eta k^2 \frac{e^{-jkr}}{4\pi r} \sin \theta \hat{\boldsymbol{\phi}}$$

$$\mathbf{H} = \underbrace{-I(\pi a^2) k^2}_{\doteq K} \frac{e^{-jkr}}{4\pi r} \sin \theta \hat{\boldsymbol{\theta}}$$

$\doteq K$ (magnetic moment)
 $\doteq IS$



radius of loop = a

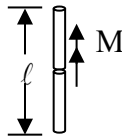
Duality (with a source)

$$\begin{array}{lll} \mathbf{J} & \rightarrow & \mathbf{M} \\ \mathbf{M} & \rightarrow & -\mathbf{J} \\ \mu & \rightarrow & \epsilon \\ \epsilon & \rightarrow & \mu \\ \mathbf{E} & \rightarrow & \mathbf{H} \\ \mathbf{H} & \rightarrow & -\mathbf{E} \\ \mathbf{A} & \rightarrow & \mathbf{F} \\ \mathbf{F} & \rightarrow & -\mathbf{A} \end{array} \quad \Rightarrow \eta \rightarrow \frac{1}{\eta} \text{ and } k \rightarrow k$$

If the loop was a true radiating magnetic dipole with magnetic moment $M\ell$ then the field solution is a direct dual of the electric dipole with the source amplitude $I\ell \rightarrow M\ell$ so:

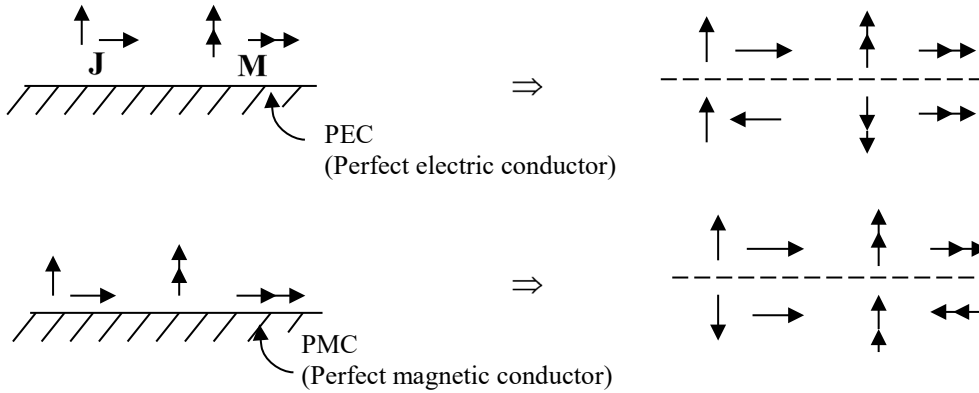
$$\mathbf{E} = -M\ell j k \frac{e^{-jkr}}{4\pi r} \sin \theta \hat{\boldsymbol{\phi}}$$

$$\mathbf{H} = M\ell \frac{j k}{\eta} \frac{e^{-jkr}}{4\pi r} \sin \theta \hat{\boldsymbol{\theta}}$$

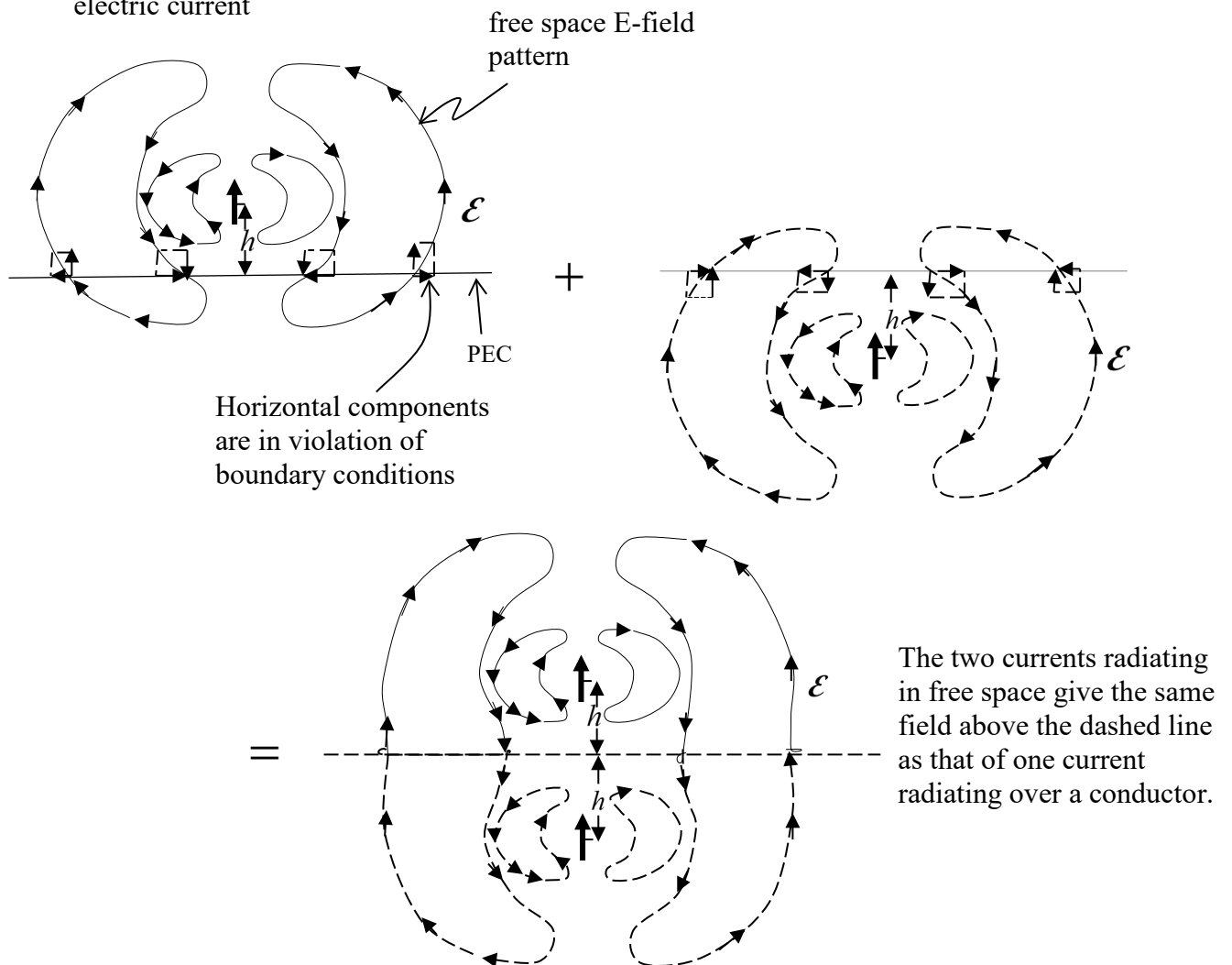


Notice that $j k \eta I S \rightarrow M\ell$ to get these equations from the electric current loop equation above. Although radiating magnetic dipoles do not exist you will see this type of approximation for the circular loop, particularly in some geophysical and remote sensing literature.

Method of Images



Example:
electric current



Note: This exercise shows that fields can also be imaged.

- Bessel Functions

A solution to the equation

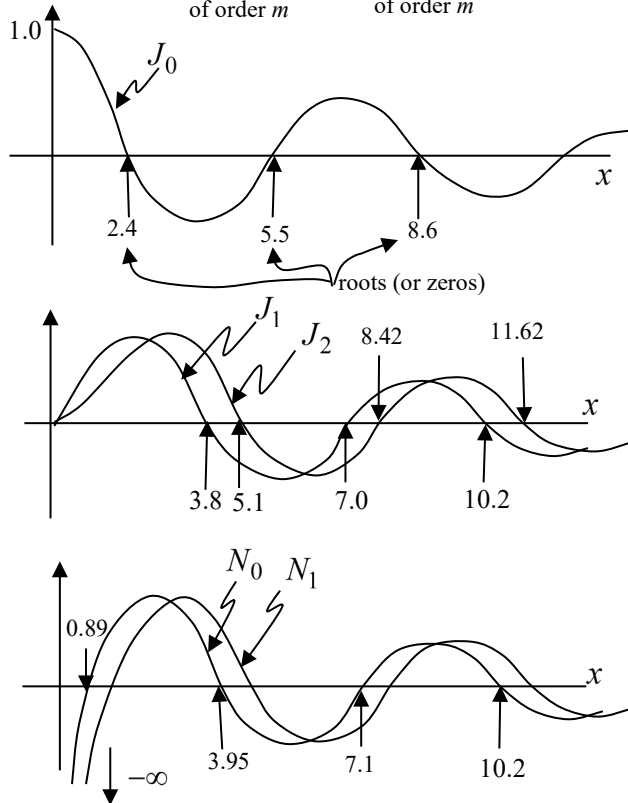
$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - m^2) y = 0$$

is

$$y = AJ_m(x) + BN_m(x) \quad \text{(or } Y_m(x))$$

Bessel function
of order m

Neumann function
of order m



Series expansion

$$J_0(x) = 1 - \frac{1}{4}x^2 + \frac{\left(\frac{1}{4}x^2\right)^2}{(2!)^2} - \dots$$

$$J_1(x) = \frac{x}{2} + \frac{x}{4} \left(-\frac{1}{4}x^2 \right) + \dots$$

$$J_m(x) = \left(\frac{x}{2} \right)^m \sum_{k=0}^{\infty} \frac{\left(-\frac{1}{4}x^2 \right)^k}{k!(m+k)!}$$

Derivative:

$$\frac{d}{dx} J_m(x) = J_{m-1}(x) - \frac{m}{x} J_m(x)$$

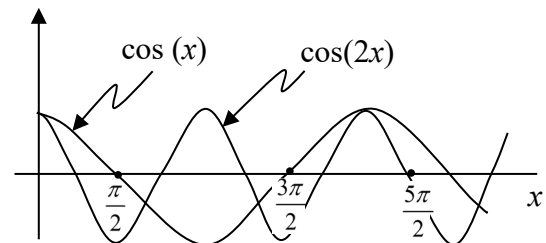
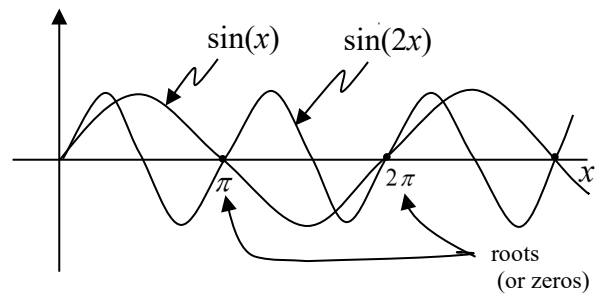
- Sine Functions

A solution to the equation

$$\frac{d^2 y}{dx^2} + n^2 y = 0$$

is

$$y = A \sin(nx) + B \cos(nx)$$



Series expansion:

$$\begin{aligned} \sin(x) &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \\ &= \sum_{k=1}^{\infty} (-1)^{k+1} \frac{x^{2k-1}}{(2k-1)!} \end{aligned}$$

Derivative

$$\frac{d}{dx} \sin(x) = \cos x$$

Vertical dipole above a PEC plane

$$r_o \simeq r - d \cos \theta$$

$$r_i \simeq r + d \cos \theta$$

$$A_z = \mu I l \frac{e^{-jk(r-d \cos \theta)}}{4\pi r} + \mu I l \frac{e^{-jk(r+d \cos \theta)}}{4\pi r}$$

$$H_\phi = \frac{E_\theta}{\eta} = \frac{j\omega A_z \sin \theta}{\eta} = \frac{jI l}{\lambda r} e^{-jkr} \cos(kd \cos \theta) \sin \theta$$

$$P_{rad} = \frac{\eta}{2} \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi/2} |H_\phi|^2 r^2 \sin \theta d\theta d\phi$$

$$= \pi \eta \int_0^{\pi/2} |H_\phi|^2 r^2 \sin \theta d\theta$$

$$= \frac{\pi \eta}{2} \int_0^\pi |H_\phi|^2 r^2 \sin \theta d\theta$$

$$P_{rad} = \frac{\pi \eta}{2} \left(\frac{I l}{\lambda r} \right)^2 r^2 \int_0^\pi \cos^2(kd \cos \theta) \sin^3 \theta d\theta$$

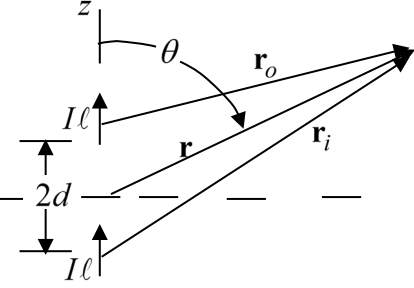
$$= \frac{\pi \eta}{2} \left(\frac{I l}{\lambda} \right)^2 \left\{ \int_0^\pi \left[\frac{1 + \cos(2kd \cos \theta)}{2} \right] \sin^3 \theta d\theta \right\}$$

$$= \frac{\pi \eta}{4} \left(\frac{I l}{\lambda} \right)^2 \left\{ \underbrace{\int_0^\pi \sin^3 \theta d\theta}_{=\frac{4}{3}} + \underbrace{\int_0^\pi \cos(2kd \cos \theta) \sin^3 \theta d\theta}_{\doteq I_2} \right\}$$

$$J_\nu(z) = \frac{\left(\frac{1}{2} z \right)^\nu}{\pi^{1/2} \Gamma\left(\nu + \frac{1}{2}\right)} \int_0^\pi \cos(z \cos \theta) \sin^{2\nu} \theta d\theta \quad ; \operatorname{Re} \nu > -\frac{1}{2}$$

9.1.20 Abramowitz & Stegun

Setting $z = 2kd$ and $\nu = \frac{3}{2}$ in this integral, I_2 above becomes,



$$\Gamma(2) = 1! \quad \text{Abromowitz (6.1.15)}$$

$$I_2 = J_{3/2}(2kd) \frac{\pi^{1/2} \Gamma\left(\frac{3}{2} + \frac{1}{2}\right)}{\left(\frac{1}{2} 2kd\right)^{3/2}} = J_{3/2}(2kd) \frac{\pi^{1/2}}{(kd)^{3/2}}$$

So

$$P_{rad} = \frac{\pi\eta}{4} \left(\frac{I\ell}{\lambda}\right)^2 \left[\frac{4}{3} + \frac{\pi^{1/2}}{(kd)^{3/2}} J_{3/2}(2kd) \right]$$

from 10.1.1 Abram & Stegun

$$j_n(z) = \sqrt{\frac{\pi}{2z}} J_{n+1/2}(z) \quad \text{where} \quad j_n(z) \equiv \text{spherical Bessel function of 1}^{\text{st}} \text{ kind}$$

$$\text{let } n=1, \quad z = 2kd$$

$$J_{3/2}(2kd) = j_1(2kd) \sqrt{\frac{2(2kd)}{\pi}}$$

$$\begin{aligned} \Rightarrow P_{rad} &= \frac{\pi\eta}{4} \left(\frac{I\ell}{\lambda}\right)^2 \left[\frac{4}{3} + \frac{\cancel{\pi^{1/2}}}{(kd)^{3/2}} \frac{2(kd)^{1/2}}{\cancel{\pi^{1/2}}} j_1(2kd) \right] \\ &= \pi\eta \left(\frac{I\ell}{\lambda}\right)^2 \left[\frac{1}{3} + \frac{1}{2(kd)} j_1(2kd) \right] \end{aligned}$$

$$\text{but } j_1(z) = \frac{\sin z}{z^2} - \frac{\cos z}{z}$$

(10.1.11)
Abram & Stegun

$$\text{let } z = 2kd$$

$$j_1(2kd) = \frac{\sin(2kd)}{(2kd)^2} - \frac{\cos(2kd)}{(2kd)}$$

$$P_{rad} = \pi\eta \left(\frac{I\ell}{\lambda}\right)^2 \left[\frac{1}{3} + \frac{\sin(2kd)}{(2kd)^3} - \frac{\cos(2kd)}{(2kd)^2} \right]$$

The reciprocity theorem

Assume two sets of current sources, each set contains an electric $\mathbf{J}$ and magnetic $\mathbf{M}$ current density.

For set (a)

$$\nabla \times \mathbf{H}^a = \mathbf{J}^a + j\omega\epsilon\mathbf{E}^a \quad (1)$$

$$-\nabla \times \mathbf{E}^a = \mathbf{M}^a + j\omega\mu\mathbf{H}^a \quad (2)$$

For set (b)

$$\nabla \times \mathbf{H}^b = \mathbf{J}^b + j\omega\epsilon\mathbf{E}^b \quad (3)$$

$$-\nabla \times \mathbf{E}^b = \mathbf{M}^b + j\omega\mu\mathbf{H}^b \quad (4)$$

Taking the dot product of (1) and $\mathbf{E}^b$ and the dot product of (4) and $\mathbf{H}^a$

$$\mathbf{E}^b \cdot \nabla \times \mathbf{H}^a = \mathbf{E}^b \cdot \mathbf{J}^a + j\omega\epsilon\mathbf{E}^b \cdot \mathbf{E}^a \quad (5)$$

$$-\mathbf{H}^a \cdot \nabla \times \mathbf{E}^b = \mathbf{H}^a \cdot \mathbf{M}^b + j\omega\mu\mathbf{H}^a \cdot \mathbf{H}^b \quad (6)$$

Adding (5) and (6) and noting that

$$\mathbf{E}^b \cdot \nabla \times \mathbf{H}^a - \mathbf{H}^a \cdot \nabla \times \mathbf{E}^b = -\nabla \cdot (\mathbf{E}^b \times \mathbf{H}^a)$$

One gets

$$-\nabla \cdot (\mathbf{E}^b \times \mathbf{H}^a) = \mathbf{E}^b \cdot \mathbf{J}^a + j\omega\epsilon\mathbf{E}^b \cdot \mathbf{E}^a + \mathbf{H}^a \cdot \mathbf{M}^b + j\omega\mu\mathbf{H}^a \cdot \mathbf{H}^b \quad (7)$$

Using the above process on equations (2) and (3) one gets

$$-\nabla \cdot (\mathbf{E}^a \times \mathbf{H}^b) = \mathbf{E}^a \cdot \mathbf{J}^b + j\omega\epsilon\mathbf{E}^a \cdot \mathbf{E}^b + \mathbf{H}^b \cdot \mathbf{M}^a + j\omega\mu\mathbf{H}^b \cdot \mathbf{H}^a \quad (8)$$

Subtract (8) from (7), integrate over a volume v and apply the divergence theorem to the left hand side. Expand on this sentence by applying the divergence theorem or give a homework problem.

$$\oint_s (\mathbf{E}^b \times \mathbf{H}^a - \mathbf{E}^a \times \mathbf{H}^b) \cdot d\mathbf{s} = \int_{vol} (\mathbf{E}^a \cdot \mathbf{J}^b - \mathbf{E}^b \cdot \mathbf{J}^a + \mathbf{H}^b \cdot \mathbf{M}^a - \mathbf{H}^a \cdot \mathbf{M}^b) dv$$

$$\oint_s (\mathbf{E}^b \times \mathbf{H}^a - \mathbf{E}^a \times \mathbf{H}^b) \cdot d\mathbf{s} = \int_{vol} (\mathbf{E}^a \cdot \mathbf{J}^b - \mathbf{E}^b \cdot \mathbf{J}^a + \mathbf{H}^b \cdot \mathbf{M}^a - \mathbf{H}^a \cdot \mathbf{M}^b) dv \quad *$$

(a) If all the currents are outside the chosen volume v , then

$$\oint_s (\mathbf{E}^b \times \mathbf{H}^a - \mathbf{E}^a \times \mathbf{H}^b) \cdot d\mathbf{s} = 0$$

This is an integral form of the Lorentz reciprocity theorem. This form can be used to prove that **two waveguide modes are orthogonal** (Collin, "Field Theory of Guided Waves," 1st ed., pp 177-178.).

(b) Another form of the reciprocity theorem is obtained when all the currents are inside the chosen volume v and $(\mathbf{E}^b \times \mathbf{H}^a - \mathbf{E}^a \times \mathbf{H}^b) \cdot d\mathbf{s} = 0$ on the surface ' s '. This can happen when the surface is a perfect conductor or the surface is at infinity (Sommerfeld radiation condition requires that the fields must be zero at infinity). In either case $*$ becomes

$$\int_{vol} (\mathbf{E}^a \cdot \mathbf{J}^b - \mathbf{E}^b \cdot \mathbf{J}^a + \mathbf{H}^b \cdot \mathbf{M}^a - \mathbf{H}^a \cdot \mathbf{M}^b) dv = 0$$

or

$$\int_{vol} (\mathbf{E}^a \cdot \mathbf{J}^b - \mathbf{H}^a \cdot \mathbf{M}^b) dv = \int_{vol} (\mathbf{E}^b \cdot \mathbf{J}^a - \mathbf{H}^b \cdot \mathbf{M}^a) dv$$

This equation says that the reaction of field ' a ' on source ' b ' is equal to the reaction of field ' b ' on source ' a '. This is known as the reaction theorem (or Rayleigh-Carson reciprocity theorem)

In a different notation $\langle a, b \rangle = \langle b, a \rangle$.

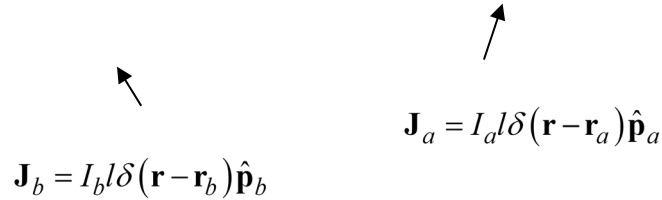
The reaction form of the reciprocity theorem

$$\int_{vol} (\mathbf{E}^a \cdot \mathbf{J}^b - \mathbf{H}^a \cdot \mathbf{M}^b) dv = \int_{vol} (\mathbf{E}^b \cdot \mathbf{J}^a - \mathbf{H}^b \cdot \mathbf{M}^a) dv$$

is very important for antenna pattern measurements:

Example/ Two currents $\mathbf{J}_a, \mathbf{J}_b$ of infinitesimal length l

$J_a = I_a l \delta(x - x_a) \delta(y - y_a) \delta(z - z_a) \doteq I_a l \delta(\mathbf{r} - \mathbf{r}_a)$, and similarly $J_b \doteq I_b l \delta(\mathbf{r} - \mathbf{r}_b)$ and polarized in the directions $\hat{\mathbf{p}}_a, \hat{\mathbf{p}}_b$.



$$\mathbf{J}_b = I_b l \delta(\mathbf{r} - \mathbf{r}_b) \hat{\mathbf{p}}_b$$

$$\mathbf{J}_a = I_a l \delta(\mathbf{r} - \mathbf{r}_a) \hat{\mathbf{p}}_a$$

From the reaction theorem

$$\int_{vol} \mathbf{E}^a \cdot \mathbf{J}^b dv = \int_{vol} \mathbf{E}^b \cdot \mathbf{J}^a dv$$

$$\mathbf{E}^a(\mathbf{r}_b) \cdot I_b l \hat{\mathbf{p}}_b = \mathbf{E}^b(\mathbf{r}_a) \cdot I_a l \hat{\mathbf{p}}_a$$

So if $I_a = I_b$

$$E_{p_b}^a(\mathbf{r}_b) = E_{p_a}^b(\mathbf{r}_a)$$

This says that when the currents are equal the component of electric field due to source 'a' in the direction of source 'b' is equal to the component of electric field due to source 'b' in the direction of source 'a'.

What this means is that the transmit and receive patterns for an antenna are the same.

An important fact that can be obtained from reciprocity is illustrated by the following example.

Example/ Suppose there is a region with some electric sources $\mathbf{J}_a, \mathbf{J}_b$ and a perfect conductor.

Let the volume in the reciprocity theorem be all of space except the PEC. Then, according to the radiation condition, on the outer surface at infinity $\mathbf{E}^a, \mathbf{E}^b, \mathbf{H}^a, \mathbf{H}^b$ are all zero. Also, on the surface of the conductor

$$\oint_s (\mathbf{E}^b \times \mathbf{H}^a - \mathbf{E}^a \times \mathbf{H}^b) \cdot d\mathbf{s} = 0$$

because $d\mathbf{s}$ is normal to the surface and $\mathbf{E}$ can only be perpendicular and $\mathbf{H}$ only tangential to a perfect conductor, so $\mathbf{E} \times \mathbf{H} \cdot d\mathbf{s} = 0$. So,

$$\oint_s (\mathbf{E}^b \times \mathbf{H}^a - \mathbf{E}^a \times \mathbf{H}^b) \cdot d\mathbf{s} = \int_{vol} (\mathbf{E}^a \cdot \mathbf{J}^b - \mathbf{E}^b \cdot \mathbf{J}^a + \mathbf{H}^b \cdot \mathbf{M}^a - \mathbf{H}^a \cdot \mathbf{M}^b) dv$$

Becomes

$$\int_{vol} \mathbf{E}^a \cdot \mathbf{J}^b dv = \int_{vol} \mathbf{E}^b \cdot \mathbf{J}^a dv$$

Now move the current source $\mathbf{J}^a$ infinitesimally close and parallel to the conductor surface. Then

$$\int \mathbf{E}^b \cdot \mathbf{J}^a dv \rightarrow \int \mathbf{E}^b \cdot \mathbf{J}_{tan}^a dv = \int E_{tan}^b J_{tan}^a dv = 0$$

conductor surface
↑
conductor surface

Current impressed on surface

because from the BC's E_{tan} to a perfect conductor is zero.

$$\therefore \int_{vol} \mathbf{E}^a \cdot \mathbf{J}^b dv = 0$$

But since no restriction has been placed on $\mathbf{J}^b$, this equation can only be true if $\mathbf{E}^a = 0$ at the position of $\mathbf{J}^b$.

- The conclusion is that an impressed electric current, a source current, an infinitesimal distance above the surface of a perfect electric conductor radiates no field.
- Notice that this example has extended the results we obtained by image theory because the surface here is finite and curved, where for image theory the surface must be infinite and flat. But, in this case the source is restricted to be very close and tangential to the surface.

